JIAYIN ZHANG

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EDUCATION

Nanjing University and University of Helsinki Sep. 2023 – Present
B.S. in Atmospheric Science
Sino-Finnish dual-degree program, Nanjing–Helsinki Institute in Atmospheric and Earth System Sciences
Overall GPA: 4.58/5.00 (91.6/100) | Rank: 3/78

Hong Kong University of Science and Technology (HKUST), Hong Kong Sep. 2025 – Dec. 2025
Undergraduate Exchange Program
Overall GPA: 3.925/4.30 (93.75/100)

Research interests: atmospheric chemistry; urban air quality; biogenic volatile organic compounds (BVOCs); machine learning; environmental data science; observation-driven atmospheric modeling.

AWARDS & HONORS

- **Zheng-Gang Scholarship for Overseas Study**, Nanjing University (Top 2%), 2025–2026.
- **Renmin Scholarship**, Nanjing University, 2025–2026.
- **Yan Baoling Scholarship for Overseas Exchange** (Excellence Award), Nanjing University, 2024–2025.
- **Outstanding Student**, Nanjing University, 2023–2024 and 2024–2025.
- **National Scholarship**, Ministry of Education of China (Top 1%), 2023–2024.
- **Outstanding Research Report**, Nanjing University Summer Social Practice Program, 2024.
- **Freshman College Outstanding Student**, Nanjing University, 2023–2024.

RESEARCH EXPERIENCE

Undergraduate Visiting Research Intern Jul. 2026 – Present
Advisor: Prof. Yuan Wang, Stanford University
Observation-constrained reconstruction of high-resolution urban isoprene emissions
- Built a unified high-resolution spatiotemporal framework for New York City by integrating TEMPO HCHO/NO_2, meteorology, vegetation, land cover, anthropogenic emissions, and urban BVOC emission estimates.
- Developed a hybrid machine-learning model to reconstruct hourly isoprene emissions using observation-constrained information and high-resolution emission priors.
- Designed temporal and spatial block validation, residual-learning diagnostics, and uncertainty analyses to evaluate reconstruction skill and generalizability.

Research Assistant Dec. 2025 – Present
Advisors: Prof. Jianzhen Yu and Prof. Lixue Cheng, Hong Kong University of Science and Technology
Regime transitions and POC/SOC partitioning of urban organic aerosols in Shanghai
- Characterized regime-dependent temporal dynamics and carbon partitioning of Shanghai urban organic aerosols from long-term observations.
- Implemented PCA–GMM clustering and transition-matrix diagnostics to quantify regime residence times, stationary distributions, and switching across pollution episodes.
- Evaluated inferred POC/SOC fractions against literature constraints and examined data-driven prediction of regime evolution.

Outputs:
- Ji, H., **Zhang, J.**, Liao, K., Cheng, L., & Yu, J. Z. Machine-Learned Chemical–Meteorological Regimes for Quantifying Secondary Organic Carbon in $PM_{2.5}$ in Shanghai and Its Long-Term Variability During 2013–2024. *Manuscript in preparation.*
- Ji, H., **Zhang, J.**, Liao, K., Cheng, L., & Yu, J. Z. Mechanistic Interpretation of Secondary Organic Carbon Variability: Regime-Specific Insights from the RAMCARB Framework and Backward Trajectory Analysis. *Manuscript in preparation.*

Undergraduate Researcher Aug. 2025 – Present

Advisors: Prof. Chao Yan and Prof. Lian Xue, Nanjing University and University of Helsinki

Boreal-forest new particle formation and multi-site particle number size distributions

- Analyzed particle number size distributions from 16 long-term ARM and EBAS stations during the Sino-Finnish Helsinki Atmospheric and Earth System Scientific Expedition, using nano-ranking with preliminary growth-rate (GR), formation-rate (J), and GDE analyses.
- Combined SMEAR II N50 observations with ENSO, NAO, and AO indices to quantify teleconnections to boreal-forest new particle formation (College Student Innovation Training Program).
- Characterized seasonal and interannual N50 controls with lagged cross-correlation, spectral analysis, joint NAO–AO composites, and radiation-stratified wavelet coherence.

Research Collaborator Jan. 2025 – Dec. 2025

Advisor: Prof. Maricar Aguilos, North Carolina State University

Hydrological stress, carbon fluxes, and tree-mortality risk in coastal and bottomland forests

- Quantified hydrological and carbon-flux drivers of tree mortality using generalized additive models, Mann–Kendall tests, and regression analyses in R.
- Established ROC-based early-warning thresholds and evaluated classification performance for tree-mortality risk prediction.

Representative outputs:

- Aguilos, M., **Zhang, J.**, Belgado, M. L., Sun, G., McNulty, S., & King, J. (2025). Determining Early Warning Thresholds to Detect Tree Mortality Risk in a Southeastern U.S. Bottomland Hardwood Wetland. *Forests*, 16(8), 1255. doi:10.3390/f16081255.
- Aguilos, M., **Zhang, J.**, Belgado, M., Sun, G., & King, J. Linking Hydrological Stress to Ghost Forest Formation: Thresholds for Tree Mortality in Coastal Ecosystems. *AmeriFlux Annual Meeting*, Tucson, USA, Oct. 2025.
- Aguilos, M. M., Sun, G., Liu, N., **Zhang, J.**, Belgado, M., McNulty, S., & King, J. S. Modeling Evapotranspiration Dynamics Across Southeastern U.S. Ecosystems Using Flux Data and Machine Learning. *AGU Annual Meeting*, New Orleans, USA, Dec. 2025.

SELECTED COMPETITIONS

First Prize, China International College Students' Innovation Competition (CICSIC) Aug. 2026

Finland Regional Competition; Project Sentinella

Selected at the Finland regional of a flagship international university innovation competition. Developed Sentinella, an integrated digital-twin system for hyperlocal-scale climate and public-health risk, and received a US$3,000 award.

Invited Presenter, United Nations Climate Change Conference (COP30) Nov. 2025

Belém, Brazil; Asian Climate–SDG Technology Innovation Hackathon Finalist

Invited to present Sentinella on the UNFCCC COP30 stage after selection as a finalist of the 2025 Asian Climate–SDG Technology Innovation Hackathon, bringing a hyperlocal urban digital-twin approach to an international climate-policy audience.

INTERNATIONAL ENGAGEMENT & LEADERSHIP

Selected Delegate, UNITAR & Nanjing University UN Immersion Programme Jul. 2026

Completed a multilateral diplomacy and global governance immersion at the Palais des Nations in Geneva, including UN system briefings and engagement with Geneva-based international organizations.

Student Ambassador, Global Training Initiative, NC State University Feb. 2025 – Present

Facilitate communication between faculty and students and support international academic engagement.

Standing Representative, Nanjing University Student Representative Congress 2024 – 2026

Serve three consecutive terms (2024, 2025, and 2026); represent student interests and contribute to university-level student affairs and campus initiatives.

Class Monitor, Nanjing–Helsinki Institute in Atmospheric and Earth System Sciences Sep. 2024 – Sep. 2025

Organize academic, social, and intercollegiate activities for undergraduate students.

COURSE COMPLETED

Selected foundation coursework: Introduction to Atmospheric Science; Fluid Dynamics; Introduction to Environmental Science; Calculus I & II; Linear Algebra; Probability Theory and Mathematical Statistics; Ordinary Differential Equations; College Physics I & II; General Chemistry; Scientific Computing I.

Research skills: Python, R, MATLAB, LaTeX; statistical modeling; machine learning; satellite and in situ data analysis; scientific visualization.

Languages: Mandarin Chinese (native); English (proficient, IELTS 7.5)